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PTO/AIA/82A (07-13)
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NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	To be assigned
Filing Date	Herewith
First Named Inventor	KO, Jae Jun
Title	MOBILE DEVICE FOR MEASURING RADON IN GROUNDWATER
Art Unit	To be assigned
Examiner Name	To be assigned
Attorney Docket Number	4016-1059

SIGNATURE of Applicant or Patent Practitioner

Signature	/Yun H. Choe/	Date (Optional)	
Name	Yun H. Choe	Registration Number	61798
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.

☑ *Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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Docket No. 4016-1059

POWER OF ATTORNEY
TO PROSECUTE APPLICATIONS BEFORE THE USPTO

I am the Applicant:

☐ Inventor or Joint Inventor

☐ Legal Representative of a Deceased or Legally Incapacitated Inventor

☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign

☐ Person Who Otherwise Shows Sufficient *Proprietary* Interest in Accordance with 35 U.S.C. § 118 and 37 CFR 1.46 (*e.g.*, a petition under 37 CFR 1.46(b)(2) was granted in the application or is being filed concurrently with this document)

I hereby revoke all previous powers of attorney given in the application identified in the attached transmittal letter.

☒ I hereby appoint Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s) and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter.

20311

Please recognize or change the correspondence address for the application to the address associated with the above-mentioned Customer Number.

APPLICANT INFORMATION

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (*e.g.*, where the applicant is a juristic entity).

Name of Applicant:	FTLAB CO., LTD.	
Applicant's Signature:		Date: 7/9 (Optional)
Name:	Jae Jun Ko	
Title:	CEO	

Note: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. Submit multiple forms for more than one signature and indicate the total number of forms below.

☒ A total of form(s) is/are submitted.

LUCAS & MERCANTI, LLP
30 BROAD STREET, 21st FLOOR
NEW YORK, NY 10004

Docket No.	4016-1059

DECLARATION (37 CFR § 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR § 1.76)

TITLE OF INVENTION	**MOBILE DEVICE FOR MEASURING RADON IN GROUNDWATER**

As the below named inventor, I hereby declare that:

This declaration is directed to:

☒ The attached application, or

☐ United States application or PCT international application number

_____on_____.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. § 1001 by fine or imprisonment of not more than five (5) years, or both.

LEGAL NAME OF INVENTOR		
Inventor Name:	KO, Jae Jun	
Signature:		Date (Optional): 7/7

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity must accompany this form. Use an additional declaration form for each additional inventor.

LUCAS & MERCANTI, LLP
30 Broad Street, 21st Floor
New York, NY 10004

Docket No.	4016-1059

DECLARATION (37 CFR § 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR § 1.76)

TITLE OF INVENTION	**MOBILE DEVICE FOR MEASURING RADON IN GROUNDWATER**

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The above-identified application was made or authorized to be made by me.

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I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. § 1001 by fine or imprisonment of not more than five (5) years, or both.

LEGAL NAME OF INVENTOR		
Inventor Name:	KIM, Young Gweon	
Signature:		Date (Optional): 2024-7-7

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity must accompany this form. Use an additional declaration form for each additional inventor.

LUCAS & MERCANTI, LLP
30 Broad Street, 21st Floor
New York, NY 10004

Docket No.	4016-1059

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I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. § 1001 by fine or imprisonment of not more than five (5) years, or both.

LEGAL NAME OF INVENTOR

Inventor Name:	SEO, Yong Jin	
Signature:		Date (Optional): 2021.7.7

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity must accompany this form. Use an additional declaration form for each additional inventor.

LUCAS & MERCANTI, LLP
30 Broad Street, 21st Floor
New York, NY 10004

Docket No.	4016-1059

DECLARATION (37 CFR § 1.63) FOR UTILITY OR DESIGN APPLICATION USING AN APPLICATION DATA SHEET (37 CFR § 1.76)

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This declaration is directed to:

☒ The attached application, or

☐ United States application or PCT international application number

_____on_____.

The above-identified application was made or authorized to be made by me.

I believe that I am the original inventor or an original joint inventor of a claimed invention in the application.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. § 1001 by fine or imprisonment of not more than five (5) years, or both.

LEGAL NAME OF INVENTOR

Inventor Name:	MAENG, Min Jae	
Signature:		Date (Optional): 2021. 7. 7.

Note: An application data sheet (PTO/SB/14 or equivalent), including naming the entire inventive entity must accompany this form. Use an additional declaration form for each additional inventor.

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PTO-1390 (01-18)
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TRANSMITTAL LETTER TO THE UNITED STATES DESIGNATED/ELECTED OFFICE (DO/EO/US) CONCERNING A SUBMISSION UNDER 35 U.S.C. 371

Attorney Docket No.	
4016-1059	
U.S. Application No. (if known, see 37 CFR 1.5)	

International Application No.	International Filing Date	Priority Date Claimed
PCT/KR2020/012682	September 21, 2020	December 27, 2019

Title of Invention
MOBILE DEVICE FOR MEASURING RADON IN GROUNDWATER

First Named Inventor
KO, Jae Jun

Applicant herewith submits to the United States Designated/Elected Office (DO/EO/US) the following items and other information.

1. ■ This is an express request to begin national examination procedures (35 U.S.C. 371(f)). NOTE: The express request under 35 U.S.C. 371(f) will not be effective unless the requirements under 35 U.S.C. 371(c)(1), (2), and (4) for payment of the basic national fee, copy of the International Application and English translation thereof (if required), and the oath or declaration of the inventor(s) have been received.
2. ■ A copy of the International Application (35 U.S.C. 371(c)(2)) is attached hereto (not required if the International Application was previously communicated by the International Bureau or was filed in the United States Receiving Office (RO/US)).
3. An English language translation of the International Application (35 U.S.C. 371(c)(2))
   a. ■ is attached hereto.
   b. ☐ has been previously submitted under 35 U.S.C. 154(d)(4).
4. An oath or declaration of the inventor(s) (35 U.S.C. 371(c)(4))
   a. ■ is attached.
   b. ☐ was previously filed in the international phase under PCT Rule 4.17(iv).

Items 5 to 8 below concern amendments made in the international phase.

PCT Article 19 and 34 amendments

5. ☐ Amendments to the claims under PCT Article 19 are attached (not required if communicated by the International Bureau) (35 U.S.C. 371(c)(3)).
6. ☐ English translation of the PCT Article 19 amendment is attached (35 U.S.C. 371(c)(3)).
7. ☐ English translation of annexes (Article 19 and/or 34 amendments only) of the International Preliminary Examination Report is attached (35 U.S.C. 371(c)(5)).

Cancellation of amendments made in the international phase

8a. ☐ Do not enter the amendment made in the international phase under PCT Article 19.

8b. ☐ Do not enter the amendment made in the international phase under PCT Article 34.

NOTE: A proper amendment made in English under Article 19 or 34 will be entered in the U.S. national phase application absent a clear instruction from applicant not to enter the amendment(s).

The following items 9 to 17 concern a document(s) or information included.

9. ■ An Information Disclosure Statement under 37 CFR 1.97 and 1.98.
10. ■ A preliminary amendment.
11. ■ An Application Data Sheet under 37 CFR 1.76.
12. ☐ A substitute specification. NOTE: A substitute specification cannot include claims. See 37 CFR 1.125(b).
13. ■ A power of attorney and/or change of address letter.
14. ☐ A computer-readable form of the sequence listing in accordance with PCT Rule 13*ter*.3 and 37 CFR 1.821-1.825 (not required if sequence listing in text format was indicated on the PCT Request as part of the International Application and the sequence listing was published as part of the international application).
15. ■ Assignment papers *(cover sheet and document(s))*. Name of Assignee: FTLAB CO., LTD.
16. ☐ 37 CFR 3.73(c) Statement *(when there is an Assignee)*.

This collection of information is required by 37 CFR 1.414 and 1.491-1.492. The information is required to obtain or retain a benefit by the public, which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 15 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Mail Stop PCT, Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

U.S. APPLN. No. (if known – see 37 CFR 1.5)	INTERNATIONAL APPLICATION No.	ATTORNEY DOCKET No.
	PCT/KR2020/012682	4016-1059

17. ☐ Other items or information:

The following fees have been submitted.	CALCULATIONS
18. ■ Basic national fee (37 CFR 1.492(a)) …….. **$300**	$ 320
19. ■ Examination fee (37 CFR 1.492(c)) • If the written opinion prepared by ISA/US or the international preliminary examination report prepared by IPEA/US indicates all claims satisfy provisions of PCT Article 33(1)-(4)…….. **$0** • All other situations …….. **$760**	$ 800
20. ■ Search fee (37 CFR 1.492(b)) • If the written opinion prepared by ISA/US or the international preliminary examination report prepared by IPEA/US indicates all claims satisfy provisions of PCT Article 33(1)-(4)…….. **$0** • Search fee (37 CFR 1.445(a)(2)) has been paid on the international application to the USPTO as an International Searching Authority ……..**$140** • International Search Report prepared by an ISA other than the US and provided to the Office or previously communicated to the US by the IB……**$520** • All other situations ……..**$660**	540 $
TOTAL OF 18, 19, and 20 =	$ 1660
☐ Additional fee for specification and drawings filed in paper over 100 sheets (excluding sequence listing in compliance with 37 CFR 1.821(c) or (e) in an electronic medium or computer program listing in an electronic medium) (37 CFR 1.492(j)). Fee for each additional 50 sheets of paper or fraction thereof …….. **$400**	0

Total Sheets	Extra Sheets	Number of each additional 50 or fraction thereof (round **up** to a whole number)	RATE	
- 100 =	/ 50 =	0	x **$400**	$

Surcharge for furnishing any of the search fee, examination fee, or the oath or declaration after the date of commencement of the national stage (37 CFR 1.492(h))……..**$140**	$

CLAIMS	NUMBER FILED	NUMBER EXTRA	RATE	
Total claims	- 20 =		x **$100**	$
Independent claims	- 3 =		x **$460**	$
MULTIPLE DEPENDENT CLAIM(S) (if applicable)			+ **$820**	$

Fee for submission of Sequence Listing text file of 300 MB to 800 MB (37 CFR 1.21(o)(1))……..**$1,000**	$
Fee for submission of Sequence Listing text file of more than 800 MB (37 CFR 1.21(o)(2))……**$10,000**	$
Processing fee for furnishing the English translation later than 30 months from the earliest claimed priority date (37 CFR 1.492(i))……..**$140 +**	$
TOTAL OF ABOVE CALCULATIONS =	$ 1660
■ **Applicant asserts small entity status.** See 37 CFR 1.27. Fees above are reduced by ½.	
☐ **Applicant certifies micro entity status.** See 37 CFR 1.29. Fees above are reduced by ¾. Applicant must attach form PTO/SB/15A or B or equivalent.	
TOTAL NATIONAL FEE =	$
Fee for recording the enclosed assignment (37 CFR 1.21(h)). The assignment must be accompanied by an appropriate cover sheet (37 CFR 3.28, 3.31)…….. **$50.00** per property **+**	$
TOTAL FEES ENCLOSED =	$ 830

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a. ☐ A check in the amount of $_____ to cover the above fees is enclosed.

b. ☐ Please charge my Deposit Account No. _____ in the amount of $_____ to cover the above fees.

c. ☒ The Director is hereby authorized to charge additional fees which may be required, or credit any overpayment, to Deposit Account No. 02-2275 as follows:

i. ☐ any required fee.

ii. ☒ any required fee except for excess claims fees required under 37 CFR 1.492(d) and (e) and multiple dependent claim fee required under 37 CFR 1.492(f).

d. ☐ Fees are to be charged to a credit card. **WARNING:** Information on this form may become public. Credit card information should not be included on this form. Provide credit card information and authorization on PTO-2038. The PTO-2038 should only be mailed or faxed to the USPTO. However, when paying the basic national fee, the PTO-2038 may NOT be faxed to the USPTO.

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NOTE: Where an appropriate time limit under 37 CFR 1.495 has not been met, a petition to revive (37 CFR 1.137(a) or (b)) must be filed and granted to restore the International Application to pending status.

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013, and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE 1: By providing this statement under 37 CFR 1.55 or 1.78, **this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA**.

NOTE 2: A U.S. national stage application may not claim priority to the international application of which it is the national phase. The filing date of a U.S. national stage application is the international filing date. See 35 U.S.C. 363.

Correspondence Address

☒ The address associated with Customer Number: 20311 **OR** ☐ Correspondence address below

Name					
Address					
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Country		Telephone			
Email					

Signature	/Yun H. Choe/	Date	2021-07-12
Name (Print/Type)	Yun H. Choe	Registration No. (Attorney/Agent)	61798

Privacy Act Statement

The **Privacy Act of 1974 (P.L. 93-579)** requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (*i.e.*, GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor:	KO, Jae Jun	Examiner:	TBA
Serial No.:	TBA	Art Unit:	TBA
Filing Date:	Concurrently	Confirmation No.:	TBA

For: **MOBILE DEVICE FOR MEASURING RADON IN GROUNDWATER**

PRELIMINARY AMENDMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

July 12, 2021

Sir:

Prior to the examination, please amend the above-identified patent application as follows:

Amendments to the Specification begin on page **2** of this paper.

Amendments to the Claims are reflected in the **Listing of Claims** which begins on page **3** of this paper.

Remarks/Arguments begin on page **8** of this paper.

CERTIFICATE OF ELECTRONIC TRANSMISSION
I hereby certify that this document is being electronically transmitted to the Commissioner for Patents via EFS-Web on **July 12, 2021.**

LUCAS & MERCANTI, LLP
BY: /Lammie Hong/
Lammie Hong

AMENDMENTS TO THE SPECIFICATION:

Please enter the following paragraph after the title on page 1:

CROSS-REFERENCE TO RELATED APPLICATION

This application is a 371 U.S. national stage of PCT/KR2020/012682, filed September 21, 2020 which claims the benefit of Korean Patent Application No. 10-2019-0176673, filed December 27, 2019, the contents of each of which are incorporated herein by reference.

AMENDMENTS TO THE CLAIMS:

This listing of claims will replace all prior versions and listings of claims in the application.

Listing of Claims:

1. (Original) A mobile device for measuring radon in groundwater, the mobile device comprising:

a main body provided to be movable by a user and constituting an overall body;

a groundwater sample chamber provided inside and outside the main body and storing a water sample collected from the groundwater;

an air pump provided inside the main body, and configured to inject air into the water sample stored in the groundwater sample chamber at a constant pressure, and generating air bubbles;

a buffer chamber provided inside the main body and configured to temporarily store the air bubbled by the air pump;

a radon measurement part provided inside the main body, and configured to suction the air temporarily stored in the buffer chamber, detect in real time alpha particles present in the buffer chamber, output a predetermined alpha particle detection signal, count for a measurement time preset based on the output alpha particle detection signal, and calculate a radon concentration measurement value;

a display provided to be exposed to an outer surface of the main body and configured to display the radon concentration measurement value calculated from the radon measurement part on a display screen; and

a control part provided inside the main body, and configured to control an operation of the air pump, the buffer

chamber, and the radon measurement part so that the air bubbled by the air pump is moved to the radon measurement part via the buffer chamber, and configured to control an operation of the display so as to quantify the radon concentration measurement value calculated from the radon measurement part and display the radon concentration measurement value on the display screen.

2. (Original) The mobile device of claim 1, further comprising:

a communication part configured to wirelessly transmit a current radon concentration measurement value calculated from the radon measurement part, wherein the control part controls an operation of the communication part so that the current radon concentration measurement value calculated from the radon measurement part is transmitted to an external terminal in real time.

3. (Original) The mobile device of claim 2, wherein the external terminal receives the current radon concentration measurement value transmitted in real time from the communication part through a specific pre-installed application service, and displays the current radon concentration measurement value on the display screen so as to allow the user to visually check, in real time, the current radon concentration measurement on the basis of the received current radon concentration measurement value.

4. (Currently Amended) The mobile device of claim 2~~ or 3~~, wherein the external terminal collects the current radon concentration measurement value, transmitted in real time from the radon measurement part during a preset measurement period, through the pre-installed specific application services,

calculates and quantifies an average value of the collected radon concentration measurement values, and displays the average value on the display screen.

5. (Original) The mobile device of claim 4, wherein the external terminal converts the current radon concentration measurement value transmitted in real time from the radon measurement part and/or a quantified radon concentration average value through the pre-installed specific application services into a database (DB) for each groundwater sample so as to be stored and managed therein.

6. (Original) The mobile device of claim 4, wherein, through the pre-installed specific application services, the external terminal generates a preset warning message of a radon risk according to the current radon concentration measurement value and/or a quantified radon concentration average value and displays the preset warning message on the display screen, when the corresponding current radon concentration measurement value transmitted in real time from the radon measurement part and/or the quantified radon concentration average value exceed a preset risk threshold value of radon concentration.

7. (Original) The mobile device of claim 1, wherein the control part collects the current radon concentration measurement value transmitted in real time from the radon measurement part during the preset measurement period, calculates and quantifies an average value of the collected radon concentration measurement values, and controls the operation of the display so as to display the average value on the display screen.

8. (Original) The mobile device of claim 7, further comprising:

a storage configured to store the current radon concentration measurement value transmitted in real time from the radon measurement part and a quantified radon concentration average value, wherein the control part controls to convert the current radon concentration measurement value transmitted in real time from the radon measurement part and/or the quantified radon concentration average value into a database (DB) for each groundwater sample, so as to be stored and managed in the storage.

9. (Original) The mobile device of claim 1, wherein the buffer chamber is provided with a plurality of partitions installed therein for increasing an inner surface area in order to maintain moisture in the air bubbled by the air pump at a constant humidity.

10. (Original) The mobile device of claim 1, wherein the radon measurement part comprises at least one radon measurement sensor using an ionization chamber method, and an internal capacity of the buffer chamber is provided in a range of 1.5 to 2 times larger than an internal capacity of the ionization chamber provided in the radon measuring part.

11. (New) The mobile device of claim 3, wherein the external terminal collects the current radon concentration measurement value, transmitted in real time from the radon measurement part during a preset measurement period, through the pre-installed specific application services, calculates and quantifies an average value of the collected radon concentration measurement values, and displays the average value on the display screen.12

12. (New) The mobile device of claim 11, wherein the external terminal converts the current radon concentration measurement value transmitted in real time from the radon measurement part and/or a quantified radon concentration average value through the pre-installed specific application services into a database (DB) for each groundwater sample so as to be stored and managed therein.

13. (New) The mobile device of claim 11, wherein, through the pre-installed specific application services, the external terminal generates a preset warning message of a radon risk according to the current radon concentration measurement value and/or a quantified radon concentration average value and displays the preset warning message on the display screen, when the corresponding current radon concentration measurement value transmitted in real time from the radon measurement part and/or the quantified radon concentration average value exceed a preset risk threshold value of radon concentration.

REMARKS

The specification has been amended to provide an incorporation by reference.

Claim 4 has been amended to remove multiple dependency and new claims 11-13 are added to conform to the amendment.

No new matter is added.

It is respectfully submitted that the present application is in all respects complete and in condition for favorable consideration.

If the Examiner has any questions regarding the claim amendments presented herein, it is requested that the Examiner contact the undersigned at the telephone number shown below.

An early and favorable action on the merits is earnestly solicited.

Respectfully submitted,

LUCAS & MERCANTI, LLP

By: /Yun H. Choe/

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UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor:	KO, Jae Jun	Examiner:	TBA
Serial No.:	TBA	Art Unit:	TBA
Filing Date:	Concurrently	Confirmation No.:	TBA
For:	**MOBILE DEVICE FOR MEASURING RADON IN GROUNDWATER**		

INFORMATION DISCLOSURE STATEMENT

Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

July 12, 2021

Sir:

Pursuant to M.P.E.P. § 609 and 37 C.F.R. §§ 1.56, 1.97-1.99, Applicants herewith submit form PTO/SB/08a, which lists references cited in International Search Report dated November 24, 2020 for International Application No. PCT/KR2020/012685, and Applicants are aware of.

This submission does not represent that a search has been made or that no better art exists. While the term "references" is used in citing the documents called to the Examiner's attention herein, Applicants do not make any admission that each or all of them are "prior art" references within the meaning of the statutory and case law.

Applicants reserve the right to contend, where appropriate, that a reference asserted against any claim of the present application is not prior art under the facts and the law.

Applicants also reserve the right to present appropriate arguments and/or evidence to establish patentability over the references, should one or more of the references be applied against the claims of the present application.

Applicants request that the Examiner independently determine those items which the Examiner would consider the most pertinent of all the references cited herein.

This Information Disclosure Statement is filed with the application and thus, within three months from the filing date. Therefore, no fee is believed to be due. If, on the other hand, it is determined that any further fees are due, or any overpayment has been made, the Assistant Commissioner is hereby authorized to debit or credit such sum to Deposit Account No. 02-2275.

It is respectfully requested that the references cited in the enclosed Forms PTO/SB/08/a be considered and made of record.

Respectfully submitted,
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CERTIFICATE OF ELECTRONIC TRANSMISSION
I hereby certify that this document is being electronically transmitted to the Commissioner for Patents via EFS-Web on **July 12, 2021**.

LUCAS & MERCANTI, LLP

BY: /Lammie Hong/
Lammie Hong

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Alexandria, VA 22313 - 1450
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ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
17/310,028	**07/12/2021 05:11:56 PM ET**	**4016-1059**

Title of Invention

MOBILE DEVICE FOR MEASURING RADON IN GROUNDWATER

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	5493	FILED BY	Lammie Hong
PATENT CENTER #	60238025	FILING DATE	-
CUSTOMER #	20311	FIRST NAMED INVENTOR	Jae Jun KO
INTL. APPLICATION #	PCT/KR2020/012682	INTL. FILING DATE	09/21/2020
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Yun Choe

Documents

TOTAL DOCUMENTS: 16

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
F1_KR10-1672846B1.PDF	13	Foreign Reference	877 KB
F2_KR10-1995927B1.PDF	30	Foreign Reference	1550 KB
F3_KR10-1194317B1.PDF	12	Foreign Reference	761 KB
F4_JP2003-194945A.PDF	8	Foreign Reference	1216 KB
F5_KR10-2015-0072742A.PDF	16	Foreign Reference	1721 KB
F6_KR10-2135694B1.PDF	19	Foreign Reference	1114 KB

N1_ISR.PDF	3	Non Patent Literature	3553 KB
4016-1059_SB08_IDS.PDF	4	Information Disclosure Statement (IDS) Form (SB08)	598 KB
4016-1059_LtrIDS_20210712.PDF	2	Letter specifying the conditions for filing under 37CFR1.97	167 KB
4016-1059_PTO1390.PDF	4	Documents submitted with 371 Applications	268 KB
4016-1059_ADS.PDF	9	Application Data Sheet	1227 KB
4016-1059_PREL.PDF	8	Preliminary Amendment	204 KB
4016-1059_Eng-APPL.PDF	38	Specification	300 KB
4016-1059_DECL.PDF	4	Oath or Declaration filed	399 KB
4016-1059_POA.PDF	2	Power of Attorney	311 KB
4016-1059-APPL-PCT-KR2020-012682.PDF	31	Specification - Not in English	588 KB

Digest

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

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ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
17/310,028	**07/12/2021 05:11:56 PM ET**	**4016-1059**

Title of Invention

MOBILE DEVICE FOR MEASURING RADON IN GROUNDWATER

Application Information

APPLICATION TYPE	Utility - U.S. National Stage under 35 USC 371	PATENT #	-
CONFIRMATION #	5493	FILED BY	Lammie Hong
PATENT CENTER #	60238025	AUTHORIZED BY	Yun Choe
CUSTOMER #	20311	FILING DATE	-
INTL. APPLICATION #	PCT/KR2020/012682	INTL. FILING DATE	09/21/2020
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Jae Jun KO

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
CARD / 4009	**E20217BH13171283**	**Lammie Hong**

FEE CODE	DESCRIPTION	ITEM PRICE($)	QUANTITY	ITEM TOTAL($)
2631	BASIC NATIONAL STAGE FEE	160.00	1	160.00
2633	NATIONAL STAGE EXAMINATION FEE - ALL OTHER SITUATIONS	400.00	1	400.00
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This is an assignment of the following rights, title and interest (check all that apply):

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☐ ***United States Patent Application Serial No.***
Date of Execution: ____ Date of Filing:

☐ ***United States Provisional Patent Application Serial No.***

☐ ***United States Patent No(s). ____***

☒ **International (PCT) Patent Application Serial No.** PCT/KR2020/012682

☐ ***Other (specify): ____***

Title of the Invention:

MOBILE DEVICE FOR MEASURING RADON IN GROUNDWATER

Inventors (assignors):

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SEO, Yong Jin	301, 12, Singiljungang-ro, Danwon-gu, Ansan-si, Gyeonggi-do, 15398 Republic of Korea
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Assignee:

Name	Address
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WHEREAS, we, the above-identified Inventors, have invented certain new and useful improvements in the Invention identified above and described in the above-identified patent application(s) and /or patent(s) (hereinafter referred to as "Invention");

And, whereas we desire to assign our above-identified rights, title and interest in the Invention to the above-identified Assignee;

NOW, this indenture witnesseth, that in consideration of the sum of ONE DOLLAR ($1.00), the receipt of which is hereby acknowledged, and for other good and valuable consideration;

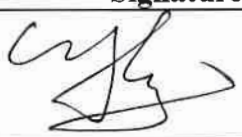
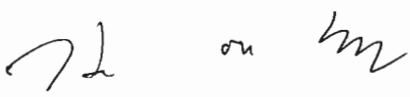
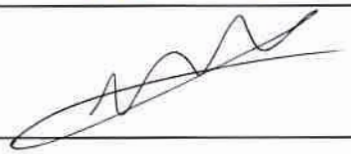
We hereby assign, sell and transfer our above-identified rights, title and interest in said Invention, said application(s) as identified above, including any divisions, continuations, and continuations-in-part thereof, and in and to any and all Letters Patent of the United States, and countries foreign thereto, which may be granted for said Invention, and in and to any and all reissues and reexaminations thereof, which may be granted or have granted for said Invention, and in and to any and all priority rights, Convention rights, and other benefits accruing or to accrue to us with respect to the filing of applications for patents or securing of patents in the United States and countries foreign thereto, unto said Assignee;

And we hereby authorize and request the Commissioner of Patents and Trademarks to issue any United States Letters Patent which may issue for said Invention to said Assignee, as assignee of the whole right, title and interest thereto;

And we further agree to sign and execute all necessary and lawful future documents, including applications for foreign patents, for filing divisions, continuations and continuation-in-part of said application for patent and/or for obtaining any reissue or reissues of any Letters Patent which may be granted for my aforesaid Invention, as the Assignee or its Designee(s) may from time to time require and prepare at its own expense.

The undersigned hereby grant(s) the firm of LUCAS & MERCANTI, LLP, located at 30 Broad Street, New York, New York, 10004, the power to insert on this assignment any further identification which may be necessary or desirable in order to comply with the rules of the United States Patent Office for recordation of this document.

Inventors' Signatures:

Name	**Signature**	**Date**
KO, Jae Jun		7/7
KIM, Young Gweon		2021-7-7
SEO, Yong Jin		2021. 7. 7
MAENG, Min Jae		2021. 7. 7.

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PATENT ASSIGNMENT COVER SHEET

Electronic Version v1.1
Stylesheet Version v1.2

SUBMISSION TYPE:	NEW ASSIGNMENT
NATURE OF CONVEYANCE:	ASSIGNMENT

CONVEYING PARTY DATA

Name	Execution Date
JAE JUN KO	07/07/2021
YOUNG GWEON KIM	07/07/2021
YONG JIN SEO	07/07/2021
MIN JAE MAENG	07/07/2021

RECEIVING PARTY DATA

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Postal Code:	15425

PROPERTY NUMBERS Total: 1

Property Type	Number
Application Number:	17310028

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ATTORNEY DOCKET NUMBER:	4016-1059
NAME OF SUBMITTER:	YUN H. CHOE
Signature:	/Yun H. Choe/
Date:	07/12/2021

Total Attachments: 2
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RECEIPT INFORMATION

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